In [7]:
```
import numpy as np
import pandas as pd

data = {
    "sepal.length": [
        5.1,4.9,4.7,4.6,5.0,5.4,4.6,5.0,4.4,4.9,
        5.4,4.8,4.8,4.3,5.8,5.7,5.4,5.1,5.7,5.1,
        5.4,5.1,4.6,5.1,4.8,5.0,5.0,5.2,5.2,4.7,
        4.8,5.4,5.2,5.5,4.9,5.0,5.5,4.9,4.4,5.1,
        5.0,4.5,4.4,5.0,5.1,4.8,5.1,4.6,5.3,5.0,
        7.0,6.4,6.9,5.5,6.5,5.7,6.3,4.9,6.6,5.2,
        5.0,5.9,6.0,6.1,5.6,6.7,5.6,5.8,6.2,5.6,
        5.9,6.1,6.3,6.1,6.4,6.6,6.8,6.7,6.0,5.7,
        5.5,5.5,5.8,6.0,5.4,6.0,6.7,6.3,5.6,5.5,
        5.5,6.1,5.8,5.0,5.6,5.7,5.7,6.2,5.1,5.7,
        6.3,5.8,7.1,6.3,6.5,7.6,4.9,7.3,6.7,7.2,
        6.5,6.4,6.8,5.7,5.8,6.4,6.5,7.7,7.7,6.0,
        6.9,5.6,7.7,6.3,6.7,7.2,6.2,6.1,6.4,7.2,
        7.4,7.9,6.4,6.3,6.1,7.7,6.3,6.4,6.0,6.9,
        6.7,6.9,5.8,6.8,6.7,6.7,6.3,6.5,6.2,5.9
    ],
    "sepal.width": [
        3.5,3.0,3.2,3.1,3.6,3.9,3.4,3.4,2.9,3.1,
        3.7,3.4,3.0,3.0,4.0,4.4,3.9,3.5,3.8,3.8,
        3.4,3.7,3.6,3.3,3.4,3.0,3.4,3.5,3.4,3.2,
        3.1,3.4,4.1,4.2,3.1,3.2,3.5,3.1,3.0,3.4,
        3.5,2.3,3.2,3.5,3.8,3.0,3.8,3.2,3.7,3.3,
        3.2,3.2,3.1,2.3,2.8,2.8,3.3,2.4,2.9,2.7,
        2.0,3.0,2.2,2.9,2.9,3.1,3.0,2.7,2.2,2.5,
        2.8,2.5,2.8,2.9,3.0,2.8,3.2,3.3,2.7,3.0,
        2.9,2.6,2.4,2.4,2.7,2.7,3.0,3.4,3.0,2.3,
        2.5,2.3,2.5,3.0,3.4,3.0,2.9,2.6,2.7,3.0,
        2.9,2.9,3.0,2.5,3.0,3.3,2.5,2.9,2.5,3.6,
        3.2,2.7,3.0,2.5,2.8,3.2,3.0,3.8,2.6,2.2,
        3.2,2.8,2.8,2.7,3.3,3.2,2.8,3.0,2.8,3.0,
        2.8,3.8,2.8,2.8,2.6,3.0,3.4,3.1,3.0,3.1,
        3.1,3.1,2.7,3.2,3.3,3.0,2.5,3.0,3.4,3.0
    ],
    "petal.length": [
        1.4,1.4,1.3,1.5,1.4,1.7,1.4,1.5,1.4,1.5,
        1.5,1.6,1.4,1.1,1.2,1.5,1.3,1.4,1.7,1.5,
        1.7,1.5,1.0,1.7,1.9,1.6,1.6,1.5,1.4,1.6,
        1.6,1.5,1.5,1.4,1.5,1.2,1.3,1.5,1.3,1.5,
        1.3,1.3,1.3,1.6,1.9,1.4,1.6,1.4,1.5,1.4,
        4.7,4.5,4.9,4.0,4.6,4.5,4.7,3.3,4.6,3.9,
        3.5,4.2,4.0,4.7,3.6,4.4,4.5,4.1,4.5,3.9,
        4.8,4.0,4.9,4.7,4.3,4.4,4.8,5.0,4.5,3.5,
        3.8,3.7,3.9,5.1,4.5,4.5,4.7,4.4,4.1,4.0,
        4.4,4.6,4.0,3.3,4.2,4.2,4.2,4.3,3.0,4.1,
        6.0,5.1,5.9,5.6,5.8,6.6,4.5,6.3,5.8,6.1,
        5.1,5.3,5.5,5.0,5.1,5.3,5.5,6.7,6.9,5.0,
        5.7,4.9,6.7,4.9,5.7,6.0,4.8,4.9,5.6,5.8,
        6.1,5.1,5.3,5.5,5.0,5.1,5.3,5.5,6.7,6.9,
        5.0,5.7,4.9,6.7,4.9,5.7,6.0,4.8,4.9,5.6
    ],
    "petal.width": [
        0.2,0.2,0.2,0.2,0.2,0.4,0.3,0.2,0.2,0.1,
        0.2,0.2,0.1,0.1,0.2,0.4,0.4,0.3,0.3,0.3,
```

```
        0.2,0.4,0.2,0.5,0.2,0.2,0.4,0.2,0.2,0.2,
        0.2,0.4,0.1,0.2,0.2,0.2,0.2,0.1,0.2,0.2,
        0.3,0.3,0.2,0.6,0.4,0.3,0.2,0.2,0.2,0.2,
        1.4,1.5,1.5,1.3,1.5,1.3,1.6,1.0,1.3,1.4,
        1.0,1.5,1.0,1.4,1.3,1.4,1.5,1.0,1.5,1.1,
        1.8,1.3,1.5,1.2,1.3,1.4,1.4,1.7,1.5,1.0,
        1.1,1.0,1.2,1.6,1.5,1.6,1.5,1.3,1.3,1.3,
        1.2,1.4,1.2,1.0,1.3,1.2,1.3,1.3,1.1,1.3,
        2.5,1.9,2.1,1.8,2.2,2.1,1.7,1.8,1.8,1.8,
        2.5,2.0,1.9,2.1,2.0,2.4,2.3,1.8,2.2,2.3,
        1.5,2.3,2.0,2.0,1.8,2.1,1.8,1.8,1.8,2.1,
        1.6,1.9,2.0,2.2,1.5,1.4,2.3,2.4,1.8,1.8,
        2.1,2.4,2.3,1.9,2.3,1.8,1.8,2.1,2.4,2.3
    ],
    "variety": ["Setosa"]*50 + ["Versicolor"]*50 + ["Virginica"]*50
}

df = pd.DataFrame(data)

df
```

Out[7]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.7	1.8	Virginica
146	6.3	2.5	6.0	1.8	Virginica
147	6.5	3.0	4.8	2.1	Virginica
148	6.2	3.4	4.9	2.4	Virginica
149	5.9	3.0	5.6	2.3	Virginica

150 rows × 5 columns

In [8]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    str    
dtypes: float64(4), str(1)
memory usage: 7.2 KB
```

In [9]:
```
df.variety.value_counts()
```

Out[9]:
```
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
```

In [10]:
```
df.head()
```

Out[10]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [11]:
```
features = df.iloc[:, :-1].to_numpy()
label = df.iloc[:, 4].to_numpy()
```

In [12]:
```
from sklearn.model_selection import train_test_split

xtrain, xtest, ytrain, ytest = train_test_split(
    features,
    label,
    test_size=0.2,
    random_state=42
)
```

In [13]:
```
from sklearn.neighbors import KNeighborsClassifier

model_KNN = KNeighborsClassifier(n_neighbors=5)
```

In [14]:
```
model_KNN.fit(xtrain, ytrain)
```

Out[14]:

▾ KNeighborsClassifier

▸ Parameters

▸ Fitted attributes

In [15]:
```
print(model_KNN.score(xtrain, ytrain))
```

```
0.9583333333333334
```

In [16]:
```
print(model_KNN.score(xtest, ytest))
```

```
1.0
```

In [17]:
```
from sklearn.metrics import confusion_matrix
```

```
confusion_matrix(label, model_KNN.predict(features))
```

Out[17]:
```
array([[50,  0,  0],
       [ 0, 49,  1],
       [ 0,  4, 46]])
```

In [18]:
```
from sklearn.metrics import classification_report

print(classification_report(label, model_KNN.predict(features)))
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.92      0.98      0.95        50
   Virginica       0.98      0.92      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

In [ ]:
```

```